ENGINEERING MATHEMATICS-II
MATLAB

Department of Science and Humanities

Fourier Transform

- Fourier transform expresses a signal (or function) $f(t)$ in the frequency domain, that is, the signal is described by a function $F(\omega)$. It is written as $F(\omega) = F\{f(t)\}$.
- In other words, the Fourier transform of a signal $f(t)$ is a signal $F(\omega)$. An alternative way of writing this is $f(t) \xrightarrow{F} F(\omega)$.
- The mathematical expression of Fourier transform of $f(t)$ is

$$F(\omega) = F\{f(t)\} = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

- It is clear that $F(\omega)$ is a complex function of ω.

Fourier Transform

- In the case of Fourier transform, $f(t)$ has to be expressed in the frequency domain. So, we replace ω by $2\pi f$. Hence,

$$F(\omega) = F\{f(t)\} = \int_{-\infty}^{\infty} f(t)e^{-i2\pi ft}dt$$

- In order to return from frequency domain to time domain, the inverse Fourier transform is applied. The inverse Fourier transform is defined by

$f(t) = F^{-1}\{F(\omega)\}$; or alternatively $F(\omega) \xrightarrow{F^{-1}} f(t)$.

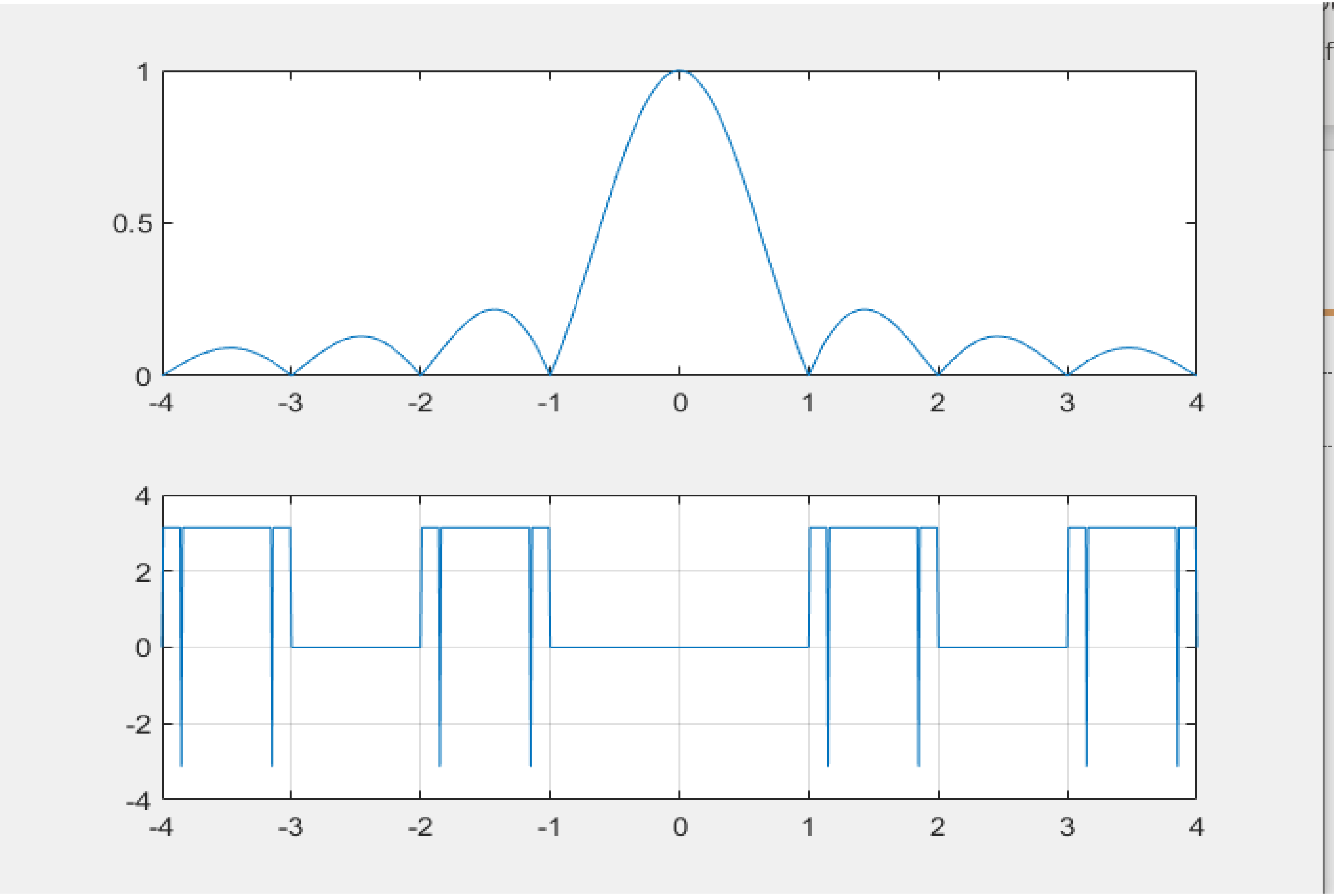
- The mathematical expression for inverse Fourier transform is

$$f(t) = F^{-1}\{F(\omega)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{i\omega t}d\omega$$

Plotting of Fourier transform of a function $f(t)$=1

```
>> clf
>> clear all
>> f=-4:0.01:4;
>> syms t
>> F=int(1*exp(-i*2*pi*f*t),t,-0.5,0.5);
>> F1=double(F);
>> subplot(211)
>> plot(f,abs(F1))
>> subplot(212)
>> plot(f,angle(F1))
>> grid on
```

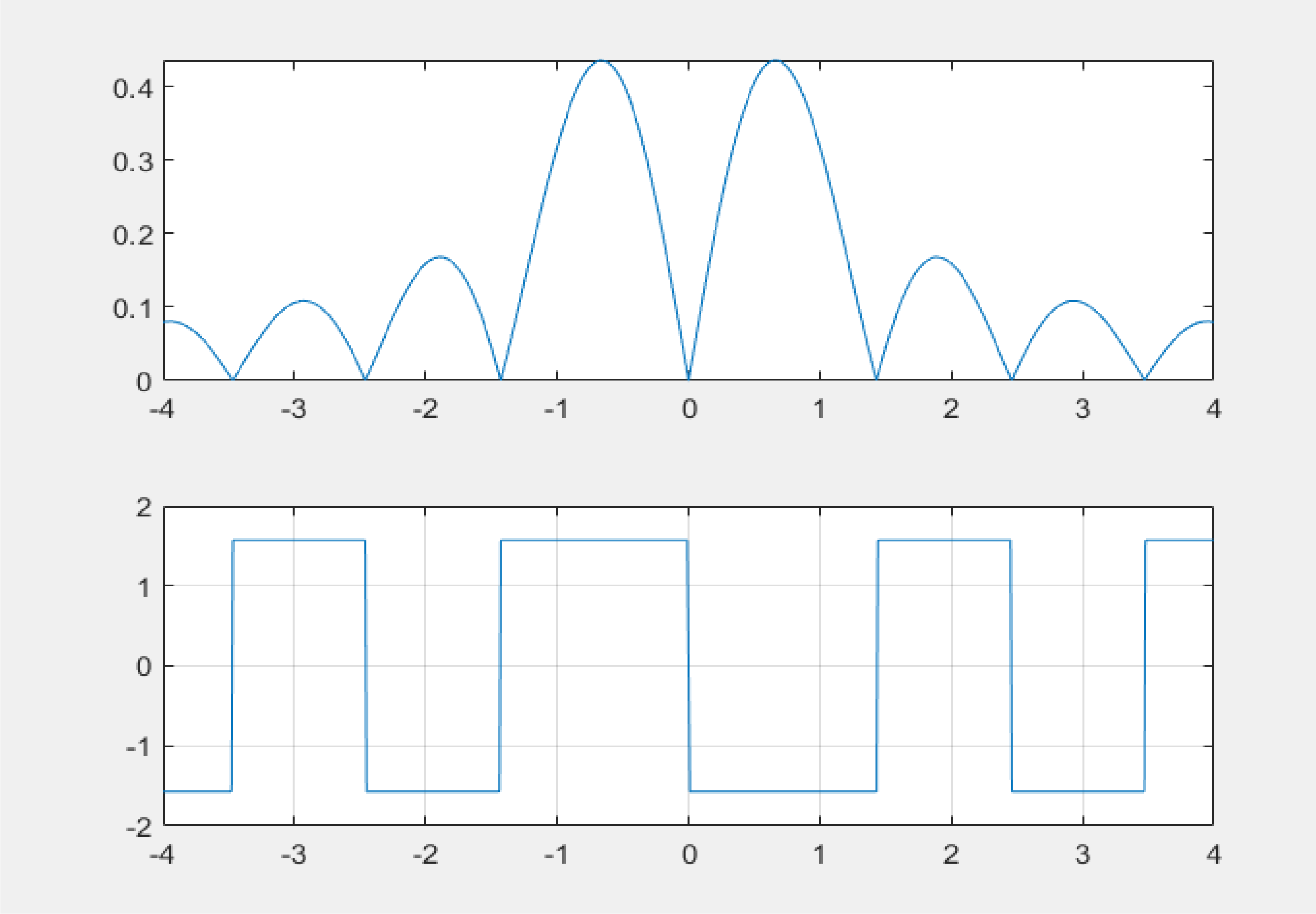
Plotting of Fourier transform of $f(t)$



Plotting of Fourier transform of $f(t) = 2t$

```
>> f=-4:0.01:4;
>> syms t
>> F=int((2*t)*exp(-i*2*pi*f*t),t,-0.5,0.5);
>> F1=double(F);
>> subplot(211)
>> plot(f,abs(F1))
>> subplot(212)
>> plot(f,angle(F1))
>> grid on
```

Plotting of Fourier transform of $f(t)$

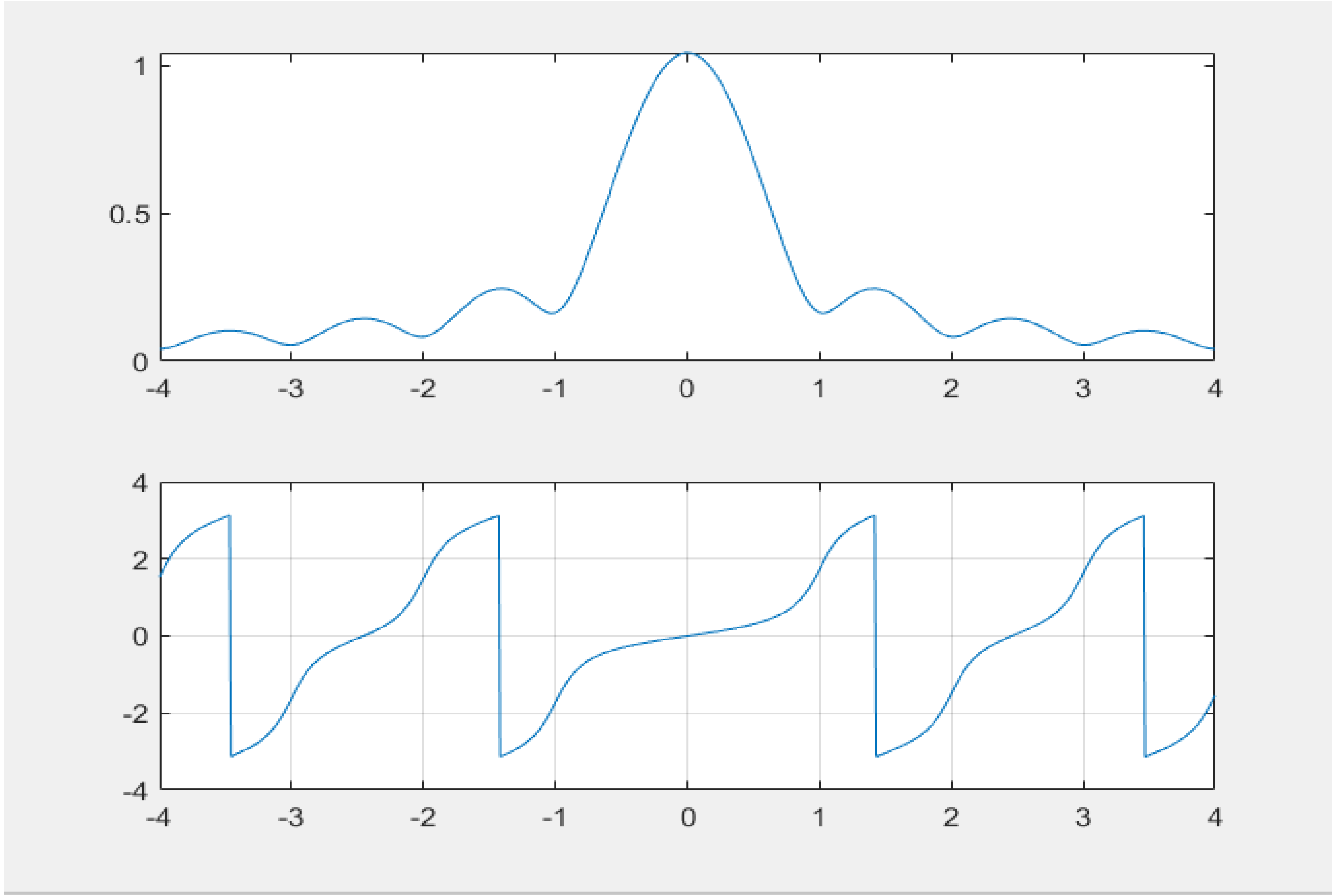


Plotting of Fourier transform of $f(t) = e^{-t}$

```
>> f=-4:0.01:4;
>> syms t
>> F=int(exp(-t)*exp(-i*2*pi*f*t),t,-0.5,0.5);
>> F1=double(F);
>> subplot(211)
>> plot(f,abs(F1))
>> subplot(212)
>> plot(f,angle(F1))
>> grid on
```

PES UNIVERSITY

Plotting of Fourier transform of $f(t)$



The command ``fourier`` and ``ifourier``

- The computation of the integral is not always a trivial method.
- In MATLAB, there is a possibility to compute the Fourier transform $F(\omega)$ of a signal $f(t)$ using the command ``fourier``.
- Correspondingly, the inverse Fourier transform is computed using the command ``ifourier``.
- While executing these two commands, time t and frequency ω must be declared as symbolic variables.

Problems:

Compute the Fourier transform of $f(t) = e^{-(t^2+x^2)}$

```
>> syms t x
>> f = exp(-t^2-x^2);
>> fourier(f)
   ans = pi^(1/2)*exp(- t^2 - w^2/4)
```

Problems:

Compute the Fourier transform of $f(t) = te^{-t^2}$

```
>> syms t x
>> f = t*exp(-t^2);
>> fourier(f)
    ans = -(w*pi^(1/2)*exp(-w^2/4)*1i)/2
```

Problems:

Compute the inverse Fourier transform of $e^{-\frac{\omega^2}{4}}$

```
>> syms w
>> F = exp(-w^2/4);
>> ifourier(F)
   ans = exp(-x^2)/pi^(1/2)
```

Note: By default, the inverse transform is in terms of x. To get the result in terms of t, we need to use the following code:

```
>> syms w t
>> F = exp(-w^2/4);
>> ifourier(F, t).
```

Here, the output is: ans=exp(-t^2)/pi^(1/2)

Problems:

Compute the Fourier transform of $f(t) = e^{-(\omega^2+a^2)}$

```
>> syms a w

>> F = exp(-w^2-a^2);

>> ifourier(F)

    ans = exp(- a^2 - x^2/4)/(2*pi^(1/2))
```

PES UNIVERSITY

THANK YOU